



DASHBOARD MANUAL

SigniQ

Panel-by-panel guide to the web dashboard: runs, signatures, the document vault, attack lab and the audit ledger.

TEAM PROMETHEUSZ · SIH26141

Quantum-secured document signatures · classical software simulation · 2026-10-02

1. Getting started

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Start the server (`cargo run -p server`) and open the printed URL — normally `http://127.0.0.1:8080` . The dashboard is a single scrolling page: a top navigation rail with ten sections (Channel · Noise vs Eve · Diagnosis · Vault · Transfer · Peers · Attack lab · Signatures · Consensus · Ledger), an editorial "folio" opener for each, and the panels beneath. On phones the rail scrolls horizontally with a hidden scrollbar; the active tab follows your scroll.

New visitors should press **Guided run** in the hero to walk the pipeline end-to-end; the **team flame** in the hero opens the six-person team dossier.

2. Reading the top bar

- **Status pill** — backend health (green/gray/red), polled continuously.
- **Guided run** — the tour button (hero).
- **Auth bar** — anonymous by default: you operate as the shared session user. Register/login switches you to a private workspace (private inbox, outbox, session key).

3. Channel — the QKD experiment

The Live Channel Monitor streams a six-state QKD simulation. Set parameters and run:

Parameter	Meaning
Key length	Qubits prepared (10k–50k typical). More qubits → tighter statistics (Hoeffding margin $\varepsilon \propto 1/\sqrt{n}$).
Threshold	Acceptance threshold on QBER, e.g. $0.15 = 15\%$. The <i>effective</i> line moves with sample size (dynamic threshold).
Eve ratio	Fraction of qubits an eavesdropper intercepts. $1.0 =$ full intercept-resend → $\text{QBER} \approx 33\%$ (measured 33.60% at 20k qubits, seed 424242).
Noise rate	Environmental bit-flip probability (fiber model). Independent of Eve — "degraded" vs "under attack".
Relay hops	Number of trusted relay nodes. Each hop costs a $(1/3)^{\text{hops}}$ yield factor.
Pace	Per-qubit delay in ms — slows the stream for narration.
Seed	Blank = fresh randomness each run; typed seed reproduces the run bit-identically .

Reading the run: the status card shows the scenario verdict, **CHSH S value** (honest channels certify at $S \approx 2.83\text{--}2.85$; classical bound 2), sifted count ($\approx 1/3$ of qubits), QBER vs the dynamic threshold, channel class (secure / degraded / under attack), and whether a key was distilled. Under attack, key distillation **aborts**. A successful secure run distills a session key the server holds; the browser never sees raw key material, only commitments.

4. Noise vs Eve — a controlled comparison

The controlled experiment separating environmental degradation from adversarial interception: same channel, controlled noise and intercept fractions, side-by-side QBER curves with the dynamic threshold overlaid. Read it as the visual argument for "degraded but honest" vs "statistically settled under attack".

5. Diagnosis — the blind challenge

A blind statistical challenge: the panel runs a hidden channel configuration and you judge honest vs attacked from statistics alone before reveal (`/api/blind-challenge/*`). Practice for the core claim — detection comes from measurement statistics, not from labels.

6. Vault — sealing and opening documents

The Quantum Document Vault seals files into opaque `.qsig` containers — a file goes in; the container comes out, and what's inside is ciphertext with a cryptographic binding, not a preview. Controls:

- **Seal** — choose a file; the server seals it under the session key (AES-256-GCM payload, HMAC-SHA256 tag, SHA-256 of the document in metadata). Optional **Shamir quorum seal** (k-of-m): the seal key is split across m officers; below k shares the document is cryptographically unrecoverable.
- **Verify** — re-checks a container: format, authenticity (QDS signature), integrity (HMAC + hash), key match (commitment). Each flag appears as an outcome flag with a note explaining any failure by name.
- **Open** — recovers the original bytes (quorum-gated when sealed with quorum).
- **Attack battery** — four tamper modes against any container; verification explains each rejection: AES-GCM tag mismatch (payload/metadata/key tampering), key-commitment mismatch (reseal under a different key).

Containers show an `audit_ref` — the Merkle audit coverage of their sealing events.

7. Transfer — streaming delivery and quorum

The transfer simulation streams a seal → send → receive → verify delivery with live SSE narration (`/api/doc/transfer`). Shamir quorum workflows (`/api/doc/quorum/*`) distribute shares, collect pledges, and unlock below-quorum-refuses / at-quorum-opens.

8. Peers — laptop-to-laptop transfer

Peer-to-peer delivery between two SigniQ instances (`/api/doc/send` with a peer URL): Alice's laptop delivers to Bob's laptop; Bob's inbox auto-checks and the verify endpoint explains the verdict. Same-server user-addressed delivery uses `to_user`. Inbox/outbox lists with delete; relay drop/claim workflow for offline delivery.

9. Attack lab — testing the document seal

The Attack Lab stages the full interception story: `capture` → `inspect` → `tamper` → `forward` → **recipient rejection** (`/api/doc/attack-theater`), each step narrated and recorded in the audit ledger. Attack modes mirror the vault battery: `tamper_bytes` , `swap_meta` , `reseal` , `truncate` .

10. Signatures — the QDS Signature Lab

The teleportation-based QDS lab (`/api/qds/*`):

- **Sign / Verify** — mint a signature (temporal trap: quantum-entropy timestamp, window sequence, chain link) and verify it (1-ACC / o-ACC / REJ verdicts with named reasons).
- **Attack battery** — six attack classes, each rejected with its reason: replay (window sequence), timestamp forgery (chain check), forgery (temporal binding), impersonation (tag binding), channel tampering (mismatch ratio), unauthorized verification (no session key material).
- **Modeled forgery probability** — $4^{-(q\lambda)}$ with the current parameters ($4^{-8} \approx 1.5 \times 10^{-5}$ at defaults; 2.9×10^{-39} at (16,4)).
- **Metrics** — verification accuracy, per-class detection rates, false-accept and false-alarm rates, operation timings (`/api/qds/metrics`).

11. Consensus — the ring and the confidence engine

Two panels share this section:

- **Consensus ring** (`/api/qds/consensus-ring`): the signature is delivered to m receivers with independent channel noise per member (a compromised member can be injected); each verifies with full trap semantics; the ring reports unlock/transfer grades. $k=3$ -of-5 tolerates one compromised member; $k=5$ -of-5 does not.
- **Statistical confidence** (`/api/stats/bounds`): Chernoff–Hoeffding intervals, the N -scaling threshold curve ("vice" that narrows with sample size), REJECT-verdict p-values with bands — including "**undersized**" when the sample cannot separate honest from attack. A detector that says "cannot decide" is a feature, not a bug.

12. Ledger — the audit trail and offline evidence

The Merkle Audit Ledger panel polls the audit ledger (8 s cadence; newest first; click a row for detail). Every seal, verification, rejection, attack, key derivation, and delivery is appended with sequence number, timestamp, actor label, accepted flag, detail, payload hash, and chain link. `GET /api/audit/verify` re-walks the chain continuously.

Offline evidence export: export a self-contained bundle and verify it offline — the browser builds a standalone verifier page (`frontend/src/auditEvidence.ts`) that checks the bundle's internal hashes, root, and inclusion proofs with no server connection. Scope, stated plainly: this proves the ledger's internal

consistency. It does not prove who published it; publisher authentication requires an expected root from an independent trusted channel. There is no key-revocation lifecycle.

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13. Accounts, P2P and the demo choreography

The classic three-laptop demo: Alice (sender), Bob (recipient), Mallory (attacker) on separate browser profiles/instances, each with an account. Alice runs the seeded QKD run, seals a document, delivers to Bob (peer URL or same-server `to_user`); Mallory captures and tampers; Bob's verification rejects with named reasons; the ledger records everything. Seeded runs make the laptops agree on the same session key without transmitting it. The deployment guide's §5 covers the k-of-m account workflow.

14. Parameters reference

Panel	Parameter	Range & effect
Channel: Key length	10k–50k qubits	More → tighter $\epsilon(n) \propto 1/\sqrt{n}$ statistics, slower run
Channel: Threshold	0.05–0.30	Acceptance line on QBER; effective line moves with n
Channel: Eve ratio	0–1	Intercept fraction; QBER → $f/3$
Channel: Noise rate	0–1	Environmental flips; "degraded" vs "under attack"
Channel: Relay hops	0–4	$(1/3)^{\text{hops}}$ yield cost
Channel: Pace	0–10 ms	Narration delay per qubit
Seed	blank or integer	Blank = fresh randomness; typed = bit-identical reproduction
Vault: Shamir k-of-m	$k \leq m$	Below k, recovery is cryptographically impossible
QDS: q (qubits), λ (repetition)	8×1 default	Forgery $P = 4^{-(q\lambda)}$: $1.5 \times 10^{-5} \rightarrow 2.9 \times 10^{-39}$

15. Interpreting results

- **QBER $\approx f/3$** — the six-state eavesdropping signature. 0% = honest; ~33% = full interception; between = partial.
- **Dynamic threshold** — noise floor + Hoeffding margin $\epsilon(n)$; narrows as samples accumulate. A verdict needs the sample *settled* (`finite_key_ok`).
- **CHSH S** — $S > 2$ certifies non-classical correlations; ≈ 2.85 honest; ≤ 2 under intercept-resend; noise above ~14.6% erases violation even without Eve.
- **Verdict bands** — airtight / decisive / significant / undersized; "undersized" means the sample cannot decide.
- **1-ACC / 0-ACC / REJ** — unconditional / non-transferable / reject. Transferability grades forwarding.
- **Key commitment** — a digest binding the container to its sealing key; mismatch = sealed under a different key.

15.1 Metrics (QDS Lab)

`/api/qds/metrics?signals=120&seed=42` (also surfaced in the lab): verification accuracy 100%, detection rate 100%, false-accept rate 0.0%, per-class detection 100% across the five classes, forgery probability $4^{-8} \approx 1.5 \times 10^{-5}$, sign 41 μs / verify 18 μs (machine-specific). These are seeded simulation measurements under the implemented test distributions — not universal guarantees.

16. Guided run and the team dossier

Guided run walks the pipeline end-to-end: channel experiment → noise-vs-eve → vault → attacks → signatures → consensus → ledger, with non-technical explanations per step. The **team flame** in the hero opens the dossier: six profiles (initials avatar when no photo), roles and duties; members can add real photos (`frontend/public/team/` , `photo: '/team/name.jpg'`), links and quotes by editing the roster in `frontend/src/components/Team.tsx` .

All quantum behavior is classical software simulation. The audit ledger proves internal consistency only. Educational prototype — not production cryptography.